

SYSTEM INTEGRATION PRACTICES

Introduction to System Integration



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Lecture Objectives

- ❑ Understand what system integration means for this course
- ❑ Understand the motivations for integrating systems
- ❑ Be aware of some very important heuristics

Introduction



Booking.com

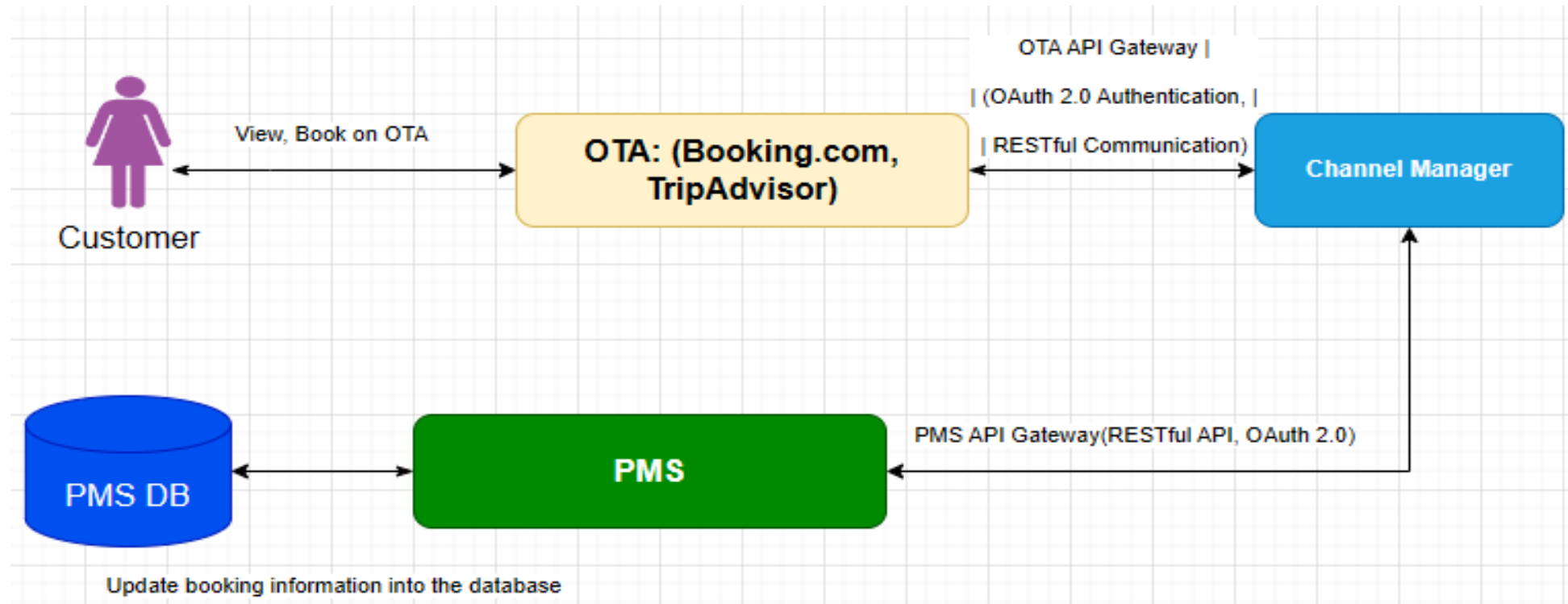


traveloka



agoda

Introduction



Integrated Hotel Management System with OTA Booking Platform

What does system integration mean?

System integration is the process of connecting multiple applications, modules, or components into a complete system that works synchronously and uniformly. The goal of this integration is to ensure that the components can effectively share data and provide users with consistent information.

Key features of system integration:

- 1.Data sharing:** Applications or modules can communicate and exchange data with each other.
- 2.Consistent information:** Users receive accurate and synchronized information across the entire system.

Examples:

- **Travel systems:** Provide users with a complete itinerary, including flight bookings, hotel reservations, and entertainment activities.
- **Enterprise accounting and project management systems:** Combine financial and project management functions into a single platform.
- **Human resource and payroll systems:** Integrate HR data with payroll functions to ensure data is synchronized and accurate.

What does system integration mean?

1. API-first Integration: is an approach to system design where **APIs (Application Programming Interfaces)** are prioritized during development, even before creating other components of the application. This ensures systems are easy to expand, integrate, and maintain.

Benefits:

- **Easy scalability:** When APIs are properly designed from the beginning, new systems can integrate without altering the existing structure.
- **Reusability:** A well-structured API can be used across various platforms (web, mobile, desktop).
- **Reduced maintenance costs:** Standardized APIs minimize the effort required for modifications and future integrations.

Example: Travel booking system: APIs are designed to connect with services like:

- **Airlines:** API provides flight schedules and ticket prices.
- **Hotels:** API offers room availability and rates.
- **Car rentals:** API lists available vehicles and rental prices.

Benefits: Third-party platforms (e.g., travel apps) can use these APIs to integrate services into their platforms, offering users a seamless experience.

What does system integration mean?

2. Event-driven Architecture: An architecture where system components communicate through **events**. When an event occurs, it triggers actions or processes without requiring direct interaction between components.

- **Benefits:**

- **Real-time processing:** Systems can react as events occur, improving user experience.
- **Scalability:** Components can be added or replaced easily without disrupting the overall system.
- **Reduced dependency:** Components don't need direct knowledge of each other, only how to handle events.

- **Example: E-commerce website:**

- **When a customer places an order:**

- An "Order Placed" event is sent to the system via **Apache Kafka** (*là một nền tảng streaming phân tán mã nguồn mở được thiết kế để xử lý dữ liệu theo thời gian thực. Nó chủ yếu được sử dụng để xây dựng các ứng dụng và hệ thống xử lý dữ liệu theo luồng lớn*).
 - Kafka distributes the event to components like:
 - **Order processing system:** *Verifies inventory and updates order status.*
 - **Notification service:** *Sends real-time notifications to the customer (email, SMS).*
 - **Shipping system:** *Initiates the shipping process.*

- **Benefits:** Quick processing, synchronization of steps, and no direct connections between systems.

What does system integration mean?

Aspect	API-first Integration	Event-driven Architecture
Goal	Design APIs for seamless integration upfront	Process and communicate systems through events
Focus	Easy integration between components	Real-time processing and reduced dependency
Common Tools	OpenAPI, Swagger	Apache Kafka, RabbitMQ
Example	Travel system with APIs for multiple services	E-commerce triggering order processing

Before you begin ...

- Why are you integrating these components/ applications/ systems?
- What is the impact to the organization's business processes?
- Are there proprietary or unique legacy systems involved?
- Has a different contractor already failed at this task?
- Who will benefit from the integration?
- What is the benefit from the integration?

Why are you integrating?

- Adaptable systems and processes
- Streamlined business processes
- Management information
- Support for electronic commerce
- Integrated security
- Replaceable components
- Reliable and recoverable systems
- Economies of scale

Adaptable system and processes

- Systems and processes are tightly coupled to computer applications
- Process changes are difficult to support with existing components
- Application knowledge is lost
- Changes are time consuming, costly, and complex
- Serverless Computing: **AWS Lambda**
- Composable Architecture

Organizations want systems that are more flexible to change

- Responsibility and control over each business function is defined and assigned
- Each business function is defined once, performed in a consistent manner
- Coupling and dependencies between business functions are minimized

- Data consistency
- Data accessibility
- Process consistency
- Exception reporting
- Historical data analysis
- Data Mesh
- Modern ETL/ELT Tools: Platforms **Fivetran** and **Airbyte**

Support for electronic commerce

- Compatibility with the systems and applications of customers and business partners
- Mechanisms for timely and reliable communication of information
- Greater security risks
- Data exchanged between your organization and an external entity create issues of trust

Integrated security

- Firewalls
- Authentication
- Authorization
- Integrity
- Confidentiality
- Nonrepudiation
- OAuth2 and OpenID Connect
- Multi-Factor Authentication (MFA)

Security integration is much more difficult than you think!

Replaceable components

- Historically, large proprietary systems could not decouple components and functionality
- Really difficult to reuse components
- Also difficult to adapt systems to different customers
- Dependent on small number of vendors
- Integration often created duplicate functionality
- Containerization (Docker, Kubernetes)
- Service Mesh (Istio)

Trend is towards finer-grained, loosely coupled components

Reliable system operations

- Minimize the risk that the system will fail
 - A safe environment with reliable power
 - Testing of equipment and software prior to operation
 - Solid change control processes
- Detect malfunctions early
 - Design and a solid testing program
- Limit the impact of failure
 - Redundancy
 - Backup and recovery designed into the system
- Multi-cloud Redundancy:
 - Use multiple cloud providers (e.g., AWS, Azure) to minimize downtime and enhance fault tolerance.

Economies of scale

- An integrated system should cost less to operate and adapt to changing needs
- Duplication should be eliminated unless it supports reliability
- Complexity should be reduced
- Economies are achieved through:
 - Standards
 - Software reuse
 - Common infrastructure
 - Consolidated systems operations
- **Improve productivity:**
 - Seamlessly integrate popular SaaS tools (e.g., Slack, Zoom)
- **Save time and reduce development costs**
 - Design reusable APIs (ex: A payment API used across multiple applications, from e-commerce to subscription services)

Technology enablers

- Middleware Technologies:** Software acting as a bridge between systems, enabling effective communication and data exchange.
- Web Technologies:** Standards and tools like HTTP, RESTful APIs, and frameworks for web development.
 - User Interface:** Intuitive and user-friendly interfaces enhance user experience in integrated systems.
 - Anywhere Access:** Ensures system accessibility from any device or location via the internet or internal networks.
- XML:** A standard for structuring and exchanging data between different systems.
 - Descriptive Tags:** Tags in XML or HTML organize and describe data for easier processing.
 - Facilitates Transformation:** Supports data and process migration across systems or formats.

Technology enablers

- **Distributed Objects:** Components spread across multiple systems, working as a unified whole:
 - **CORBA:** Enables communication between applications in different programming languages.
 - **COM+:** A Microsoft technology for integrating applications and services.
 - **Enterprise Java Beans (EJB):** A Java framework for enterprise application development.
- **GraphQL:** A query language that fetches precise data, reducing unnecessary data transfers.
- **Edge Computing:** Processes data at the network edge, minimizing latency for IoT and real-time applications.

What is Enterprise integration (EI)?

“The strategic consideration of the process, methods, tools and technologies associated with achieving interoperability between IT applications both within and external to the enterprise to enable collaborative business processes”

(Enterprise Architecture and Integration: Methods, Implementation, and Technologies, 2007)

What is Enterprise integration (EI)?

Business Integration is not merely about technology integration; it also involves aligning business processes across IT applications. Key aspects include:

1. Business Process Integration:

- Ensures workflows within the organization are synchronized and tightly connected through IT applications, improving efficiency and reducing errors.
- *VD: Một công ty bán lẻ tích hợp hệ thống quản lý kho với nền tảng thương mại điện tử. Khi khách hàng đặt hàng trực tuyến, hệ thống sẽ tự động cập nhật tồn kho, tạo yêu cầu vận chuyển và gửi thông báo đơn hàng đến cả kho hàng và khách hàng.*

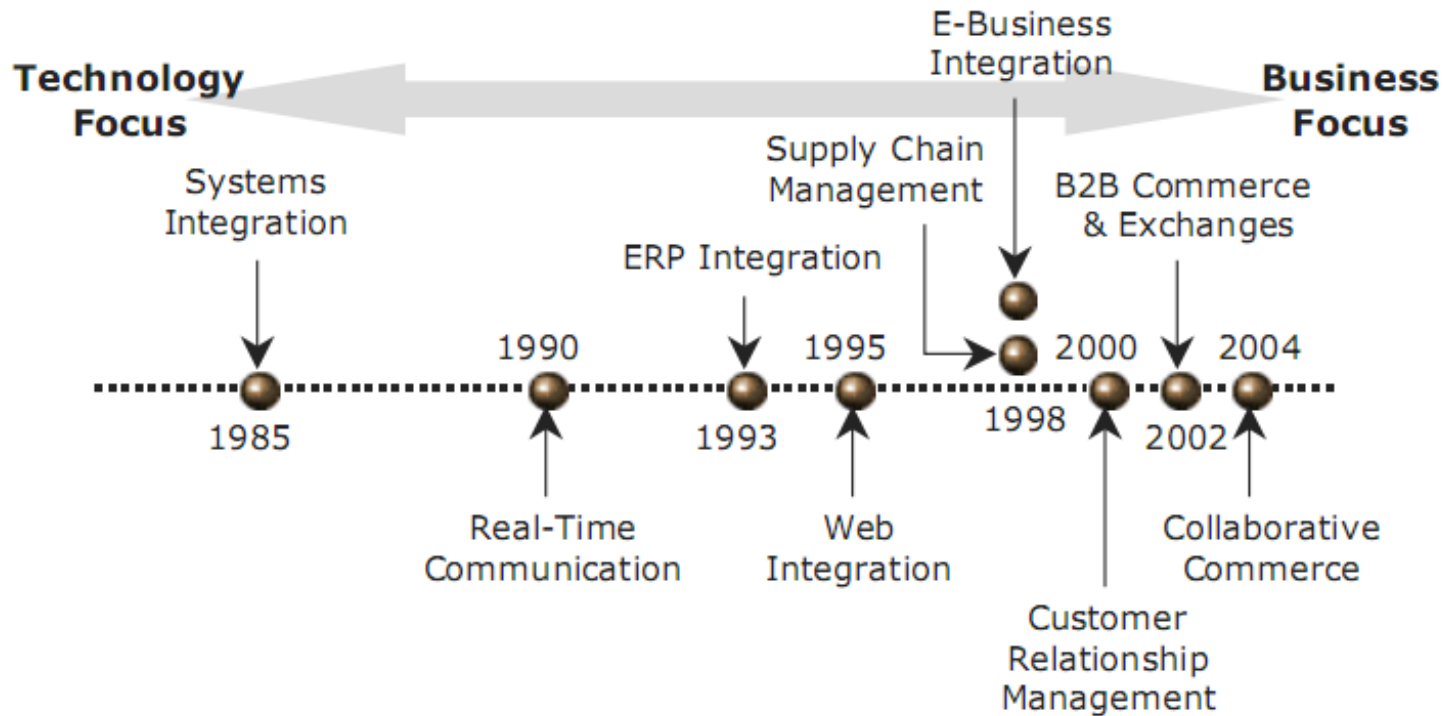
2. AI-powered Integration:

- Leverages AI tools like **IBM Watson** to assist with data analysis, process automation, and intelligent system integration.
- *VD Một ngân hàng sử dụng IBM Watson để tích hợp hệ thống hỗ trợ khách hàng. AI phân tích các câu hỏi của khách hàng, tự động trả lời các câu hỏi đơn giản và chuyển những vấn đề phức tạp đến đúng bộ phận, giúp cải thiện thời gian phản hồi và sự hài lòng của khách hàng.*

3. Hybrid Integration:

- Combines on-premises systems with cloud-based solutions to meet modern business needs while retaining legacy system functionality.
- *VD Một Cty logistics kết hợp hệ thống quản lý kho tại chỗ (on-premises) với công cụ tối ưu hóa lộ trình dựa trên đám mây. Sự tích hợp này cho phép công ty vừa giữ được quyền kiểm soát dữ liệu kho nhạy cảm vừa tận dụng công cụ đám mây để cải thiện hiệu quả giao hàng theo thời gian thực.*

EI: from a technology focus to a business focus



(Enterprise Architecture and Integration: Methods, Implementation, and Technologies, 2007)

2004 - 2010:

- Cloud Computing
- Mobile Integration
- Service-Oriented Architecture (SOA): Transition from traditional software-based integration models to service-oriented architecture, where business functions are delivered as independent services.

2010 - 2020:

- Big Data Integration
- API Economy
- Artificial Intelligence (AI)
- Internet of Things (IoT)

2020 - 2024:

- Edge Computing
- Blockchain Integration
- Hybrid Cloud and Multi-Cloud
- Low-Code/No-Code Platforms
- 5G Technology

Benefits of Business focus

- Better business responsiveness
- Improved customer support and service
- Automation of business processes
- Shortened processing cycles
- Minimized processing errors
- Enhanced partner relationships
- Improved supply chain management
- Faster introduction of new services

Challenge in system integration

- Technical challenges
- Management challenges

Technical challenges

- **Independent Design**

- Initial IT systems were designed independently.
- IT systems did not provide APIs (Application Programming Interfaces) or external interfaces.

- **Data Models**

- Applications were packaged with data models that could not be shared with other IT systems.
- No access to or sharing of data through interfaces.

Technical challenges (cont')

Heterogeneous Technology

- Different IT systems use various platforms, applications, and programming languages.
- Presence of multiple distributed models, e.g., COM, EJB, and CORBA (Common Object Request Broker Architecture).

System Engineering

- Legacy IT systems used outdated technology with no support, and original designers are no longer available, with poor documentation.
- Compatibility issues between old and new technologies (e.g., web-based).

Technical challenges (cont')

Lack of Interfaces

- Many IT systems offer limited or no interfaces or APIs for other applications to access internal functions.
- Interfaces or APIs are restricted to programming languages such as C, C++, etc.

Semantic Inconsistency

- Differences in data semantics across IT systems complicate information exchange.
- Many business partners use proprietary business terminologies.

Technical challenges (cont')

Unclear Business Processes

- Each organization adopts different data and documentation standards.
- Does not comply with common or current standards, such as XML or web services.

Management Challenges

New Collaboration

- Work extends beyond a single organization's boundaries.
- Collaboration with external partners introduces differences in policies and organizational cultures.

Integration Project Scope

- Satisfy the business integration needs of multiple stakeholders.
- Define a feasible project scope with limited resources.
- Assess and estimate the resources and time required for a complex project.

Management Challenges

Ongoing Support

- Continuous support from management throughout the entire system integration lifecycle.

Data Ownership

- Resolve data ownership issues when data spans multiple organizations.
- Clearly define who has the authority to modify data.

Time Constraints

- System integration solutions must be implemented quickly to meet business demands.

Cost Constraints

- System integration solutions must be financially efficient.
- Stay within the allowable budget.

Management Challenges

System Transition

- Keep IT applications running during integration.
- Implement system transition plans with minimal downtime.

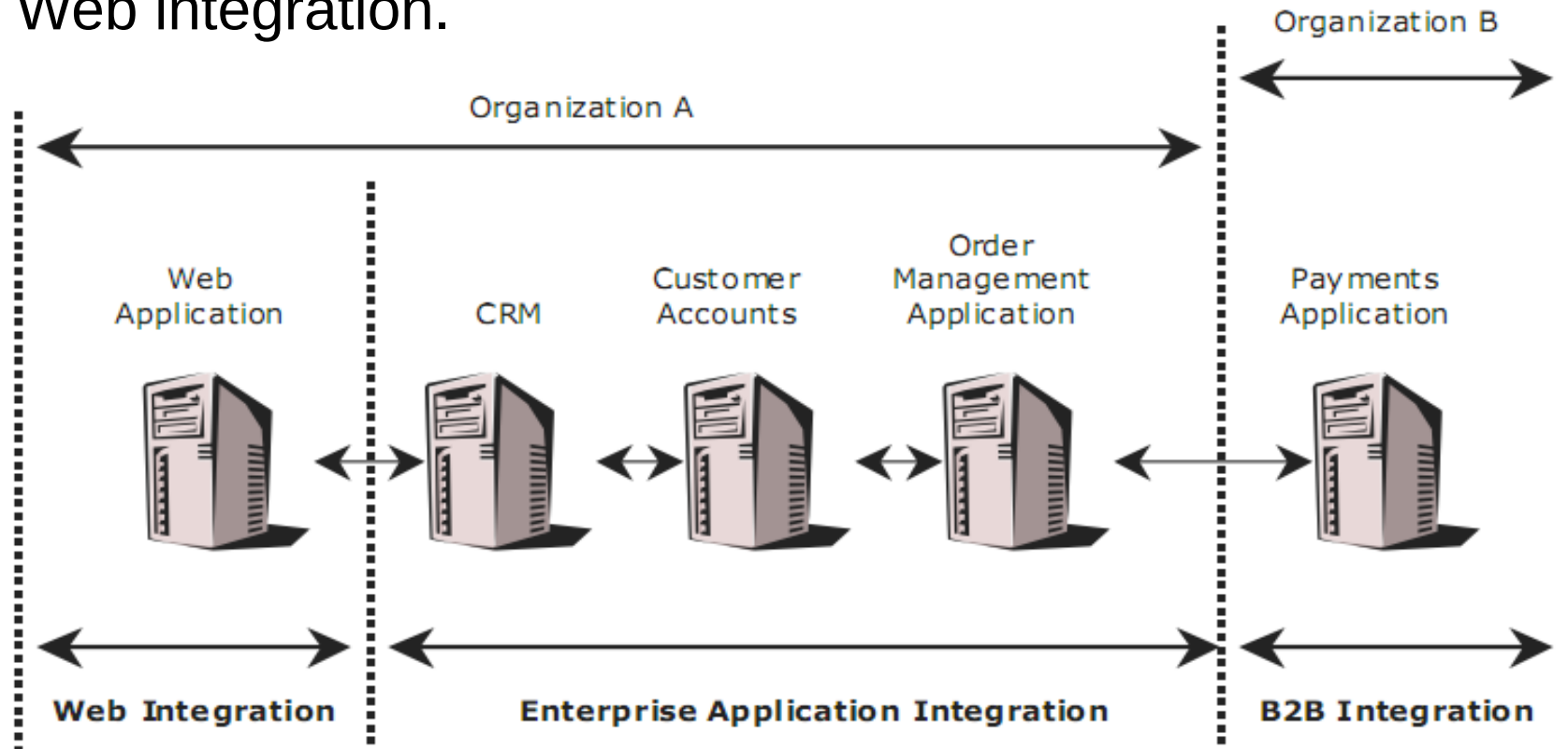
Expert Opinion

- Utilize internal or external system integration experts.

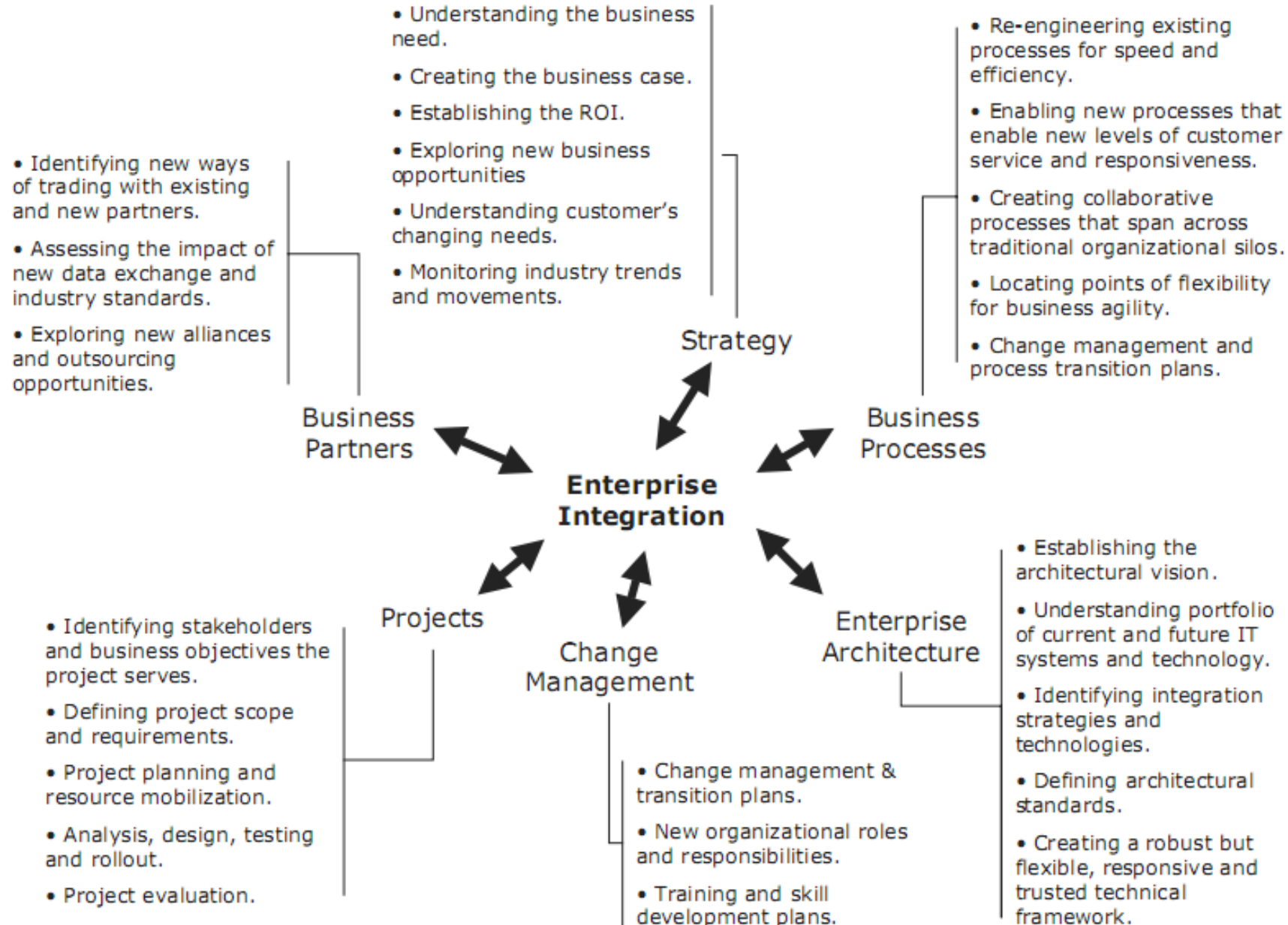
Integration Strategy

Most system integration projects fall into one of the following three categories:

- Enterprise application integration(EAI)
- Business-to-Business integration (B2Bi)
- Web integration.

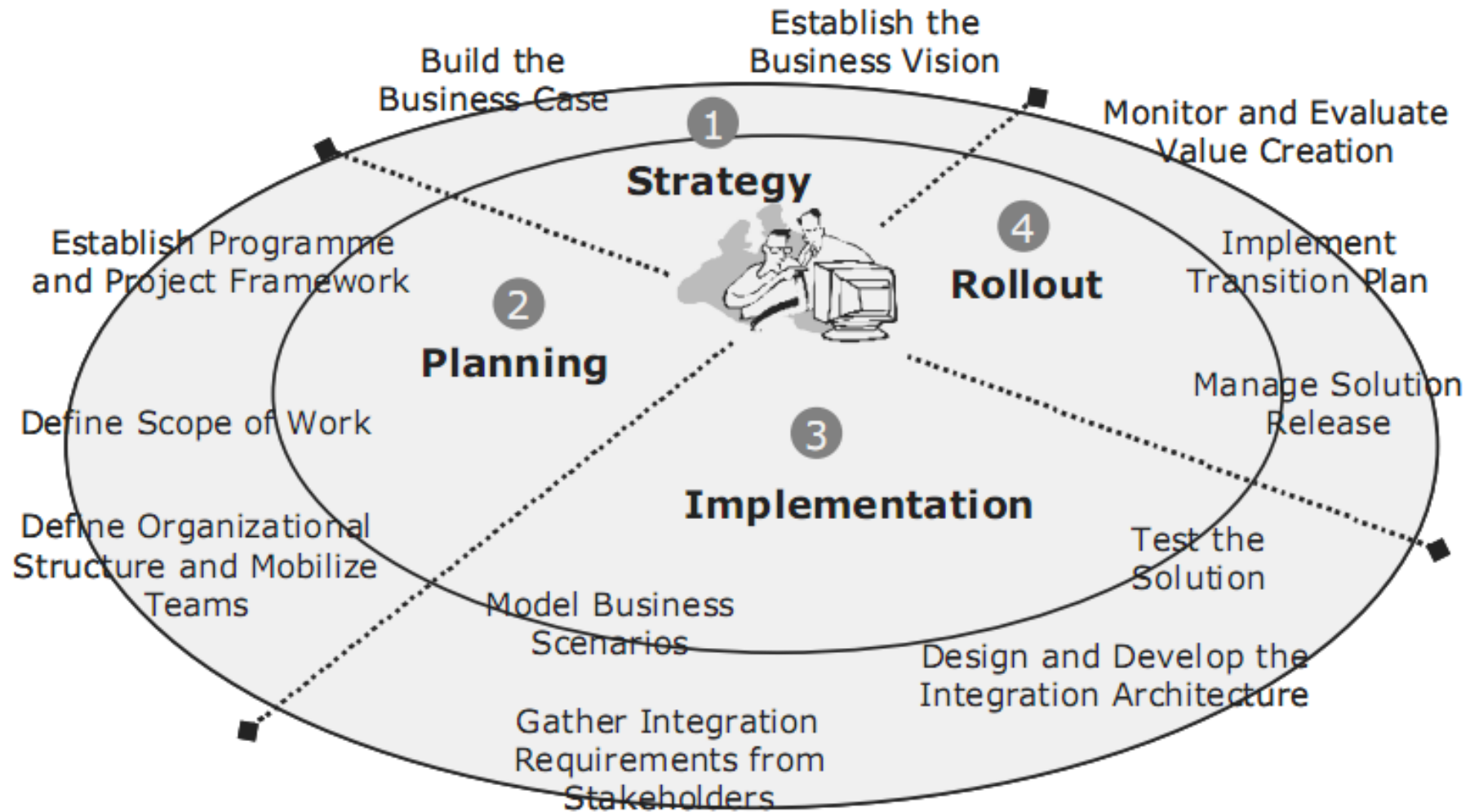


Integration Strategy



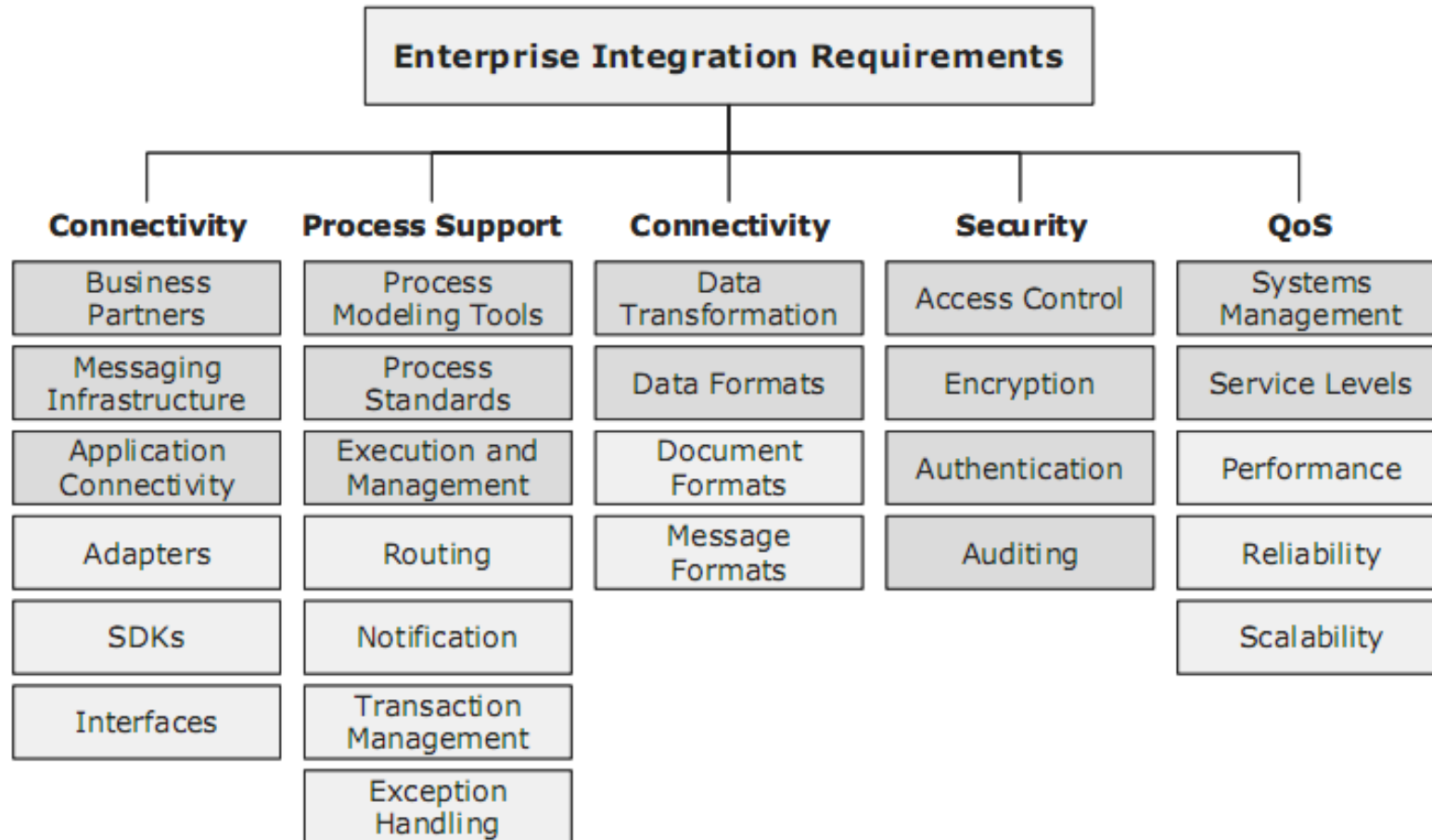
A strategic framework for enterprise integration

Integration Strategy



Model for program management

Integration Strategy



Enterprise integration requirements

The value of integration heuristics

1. Limited experience of system engineers:

- System engineers typically work on only a few complex systems throughout their careers. This limits their ability to accumulate and apply practical experience across diverse projects or types of systems.

2. Lack of standardized rules and formulas:

- Systems engineering has not yet developed the rules and formulas that form the foundation of other engineering disciplines (e.g., mechanical or electrical engineering). As a result, heuristics serve as practical guidelines to address ambiguous or unpredictable situations.

3. Heuristics are often violated:

- Even though heuristics may seem obvious and logical, system engineers frequently violate them during the design and development process. This can happen due to time constraints, insufficient information, or the overwhelming complexity of the project.

Heuristic 1

Don't assume that the original statement of the problem is necessarily the best, or even the right one.

Heuristic 2

Build and maintain options as long as possible in the design and implementation of complex systems. You will need them.

Heuristic 3

Simplify, Simplify, Simplify

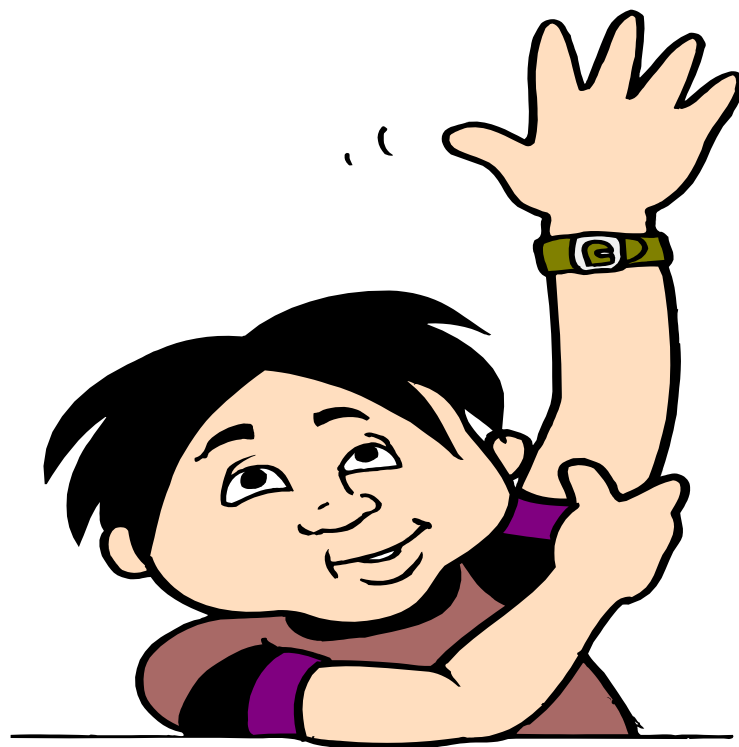
Heuristic 4

In partitioning, choose the elements so that they are as independent as possible; that is, elements with low external complexity and high internal complexity.

Summary

- Before you start thinking about integration, make sure that the purpose is clear and feasible
- Clearly understand the goals of the integration
- Good systems engineers use heuristics to guide them in design of a system
- Don't assume that the customer knows exactly what they want
- Keep the integration simple
- Keep your options open
- Keep components independent

QA



The end

THANK YOU